

2.REQUIREMENT ANALYSIS PHASE

The Requirement Analysis Phase focuses on identifying the functional and non-functional requirements of the FitBuddy application. The requirements were analyzed based on the objectives established during the brainstorming phase.

2.1 Functional Requirements

The system should allow users to enter their personal fitness information. The required information includes:

- Username.
- User ID.
- Age.
- Weight.
- Fitness goal.
- Workout intensity.

The system should validate the information entered by the user before processing it.

After receiving valid information, the application should generate a personalized seven-day workout plan using Gemini AI. Each workout plan should contain appropriate daily activities, warm-up guidance, exercises, repetitions or duration, rest periods, and cooldown guidance.

The application should also generate a nutrition or recovery tip for the user.

Another important functional requirement is the feedback system. Users should be able to provide feedback about their generated plan. FitBuddy should process the feedback and generate an updated workout plan.

The system should store user information and generated plans in a database. The administrator should be able to view registered users and their generated plans through a protected administrative interface.

The project also requires API endpoints so that the AI functionality can be accessed programmatically.

2.2 Non-Functional Requirements

The system should be:

- Easy to use.
- Responsive.

- Reliable.
- Maintainable.
- Secure enough for a demonstration project.
- Fast in processing user requests.

3 Scalable for future development.

- Compatible with modern web browsers.

2.3 Hardware Requirements

A basic development computer is sufficient:

- Processor: Intel Core i3 or equivalent and above.
- RAM: Minimum 4 GB, recommended 8 GB.
- Storage: At least 2 GB free space.
- Internet connection for Gemini API access.

2.4 Software Requirements

The major software requirements are:

- Python 3.11.
- Visual Studio Code.
- FastAPI.
- Uvicorn.
- Jinja2.
- SQLAlchemy.
- SQLite.
- Pydantic.
- Google Gemini API.
- HTML5.
- CSS3.
- JavaScript where required.

The requirement analysis phase ensures that the project has clearly defined objectives before implementation begins.

